



TECHNICAL APPROACH



Layer	Technology	What it's for	Quantum Simulator	Qiskit Aer	Running circuits on classical hardware
OS	Ubuntu Linux	Development environment	Numerical Computing	NumPy	Vector/matrix math
Frontend	React + Vite	Dashboard UI			
3D Visualization	Three.js + React Three Fiber	Bloch sphere visualization	Scientific Computing	SciPy	Statistical calculations
Charts	Recharts	Displaying detection stats	Data Validation	Pydantic	Validating API data
Backend API	FastAPI	Serving detection results	Testing	pytest, Vitest, React Testing Library	Backend + frontend testing
Backend Language	Python 3.14	Core logic	API Communication	REST + JSON	Frontend-backend communication
Quantum Framework	Qiskit 2.5.x	Building quantum states/circuits			
			Version Control	Git + GitHub	Code management